WORKED SOLUTION

STEP III 2024, Question 10

Mechanics • Toppling and Sliding

STEP III 2024, Question 10

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2024 STEP papers (a zip file):

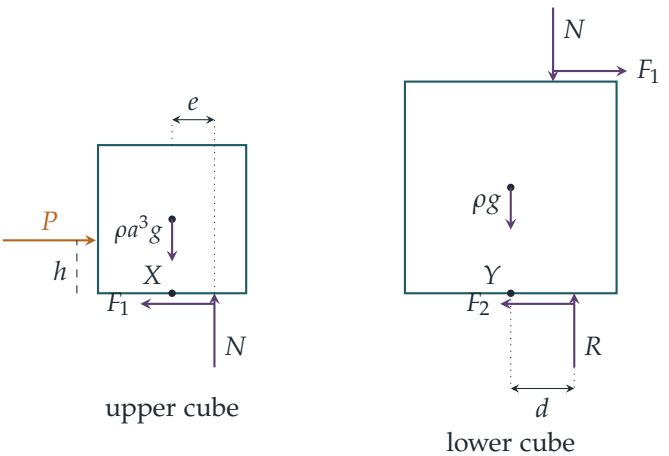
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The key idea

Every part is a comparison between four ways in which equilibrium can fail: the upper cube slides, the lower cube slides, the upper cube topples, or the two cubes topple together. Each of these happens at its own critical value of P, and whichever critical value is smallest wins as P grows from zero. Taking moments about the right point, Y for the whole system and X for the upper cube alone, gives the positions of the normal reactions in part (i) with very little work, and toppling begins when one of these reactions reaches the edge of the face it acts on.

Solution

We model the cubes as rigid bodies. The lower cube has mass ρ and the upper cube has mass ρa^3. Take P to act horizontally to the right, at height h above the top face of the lower cube, so at height $1 + h$ above the ground. Since P is horizontal, its moment about any point depends only on its height, not on where along the face it acts.



Here N and F_1 are the normal reaction and friction between the cubes (shown on each cube by Newton's third law), acting at distance e from X; R and F_2 are the normal reaction and friction from the ground, acting at distance d from Y.

Part (i)

Whole system. The only external forces are P, the two weights, R and F_2. Resolving,

$$F_2 = P, \qquad R = \rho g + \rho a^3 g = (1 + a^3)\rho g.$$

Take moments about Y. The weights both act along the vertical line through Y (each centre of mass is directly above Y), and F_2 acts along the ground through Y, so none of these has a moment. That leaves P, at height $1 + h$, balanced by R at distance d:

$$R\,d = P(1 + h) \quad \Longrightarrow \quad d = \frac{P(1 + h)}{(1 + a^3)\rho g},$$

measured from Y in the direction of P. This is the required result.

Upper cube alone. Resolving gives $F_1 = P$ and $N = \rho a^3 g$. Taking moments about X, the weight and F_1 have no moment, so $Ne = Ph$ and

$$e = \frac{Ph}{\rho a^3 g},$$

again measured from X in the direction of P.

Part (ii)

While nothing has moved, the friction forces are $F_1 = P$ between the cubes and $F_2 = P$ at the ground. Slipping happens at a contact once the friction needed there exceeds μ times the normal reaction, so

$$\text{upper cube slips on lower when } P > \mu\rho a^3 g,$$
$$\text{lower cube slips on ground when } P > \mu(1 + a^3)\rho g.$$

Since $a^3 < 1 + a^3$, the first threshold is always strictly smaller. As P increases from zero (and if no toppling intervenes), the upper cube reaches its limit first and starts to slide while the ground friction is still well within its limit. Once the upper cube is sliding, the friction it exerts on the lower cube is $\mu\rho a^3 g$, still less than the $\mu(1 + a^3)\rho g$ that the ground can supply, so the lower cube stays put. So equilibrium is broken by the upper cube slipping, not by the lower cube slipping.

Part (iii)

A body on a flat face is about to topple when the normal reaction on it reaches the edge of that face. The upper cube's base has half-width $\frac{1}{2}a$, and the lower cube's base has

half-width $\frac{1}{2}$. So, measuring every force in units of ρg, the four critical values of P are

$$P_{\text{slip}} = \mu a^3 \text{ (upper on lower)}, \qquad P_{\text{ground}} = \mu(1+a^3),$$
$$P_{\text{upper}} = \frac{a^4}{2h} \text{ (from } e = \tfrac{1}{2}a\text{)}, \qquad P_{\text{both}} = \frac{1+a^3}{2(1+h)} \text{ (from } d = \tfrac{1}{2}\text{)}.$$

Toppling of the lower cube carries the upper one with it, which is why $d = \frac{1}{2}$ means the two cubes topple together.

Now put $a = 1$, so $P_{\text{slip}} = \mu$, $P_{\text{upper}} = \frac{1}{2h}$ and $P_{\text{both}} = \frac{1}{1+h}$. By part (ii) we can ignore P_{ground}.

Ask yourself Do we need to worry about the upper cube toppling on its own?

No, but it must be checked. Since $h < a = 1$ we have $2h < 1 + h$, so

$$P_{\text{upper}} = \frac{1}{2h} > \frac{1}{1+h} = P_{\text{both}}.$$

The two cubes would topple together before the upper cube could topple alone. So the contest is only between $P_{\text{slip}} = \mu$ and $P_{\text{both}} = 1/(1+h)$:

- if $\mu(1+h) < 1$, then $\mu < \frac{1}{1+h}$, the slipping threshold is reached first, and the upper cube slips on the lower one;
- if $\mu(1+h) > 1$, then $\frac{1}{1+h} < \mu$, the toppling threshold is reached first, and the two cubes topple together.

Part (iv)

If no slipping occurs, the contest is between P_{upper} and P_{both}. The upper cube topples first exactly when

$$\frac{a^4}{2h} < \frac{1+a^3}{2(1+h)} \iff a^4(1+h) < h(1+a^3) \iff a^4 < h\big(1+a^3-a^4\big) = h\big(1+a^3(1-a)\big),$$

where we multiplied through by the positive quantity $2h(1+h)$. This is precisely the given condition, so under it equilibrium is broken by the upper cube toppling.

Part (v)

With $a = \frac{1}{2}$ we need three things.

- The condition of (iv): $h\big(1 + \frac{1}{8}\cdot\frac{1}{2}\big) > \frac{1}{16}$, that is, $\frac{17}{16}h > \frac{1}{16}$, so $h > \frac{1}{17}$.
- The setup requires $h < a = \frac{1}{2}$.
- No slipping before toppling: $P_{\text{upper}} < P_{\text{slip}}$, that is, $\dfrac{1/16}{2h} < \dfrac{\mu}{8}$, or $\mu > \dfrac{1}{4h}$. (Slipping on the ground is never first, by (ii).) Since we also need $\mu < 1$, this requires $h > \frac{1}{4}$.

So choose $h = \frac{2}{5}$, which satisfies $\frac{1}{4} < h < \frac{1}{2}$ and $h > \frac{1}{17}$. Then we need $\frac{5}{8} < \mu < 1$; take $\mu = \frac{3}{4}$. The critical values (in units of ρg) are then

$$P_{\text{upper}} = \tfrac{5}{64} \approx 0.078, \quad P_{\text{slip}} = \tfrac{3}{32} \approx 0.094, \quad P_{\text{both}} = \tfrac{45}{112} \approx 0.402, \quad P_{\text{ground}} = \tfrac{27}{32} \approx 0.844.$$

The smallest is P_{upper}, so as P grows the upper cube topples before anything slips: the situation of (iv) does occur.

Checks

- As $P \to 0$, both d and e tend to 0: with no push, each reaction acts directly below the centre of mass, as it should.
- Taking moments for the lower cube alone (its weight, R and F_2 at distance d, and N and F_1 reversed at distance e from X) gives $Rd = Ne + F_1 \cdot 1$, which is $P(1+h) = Ph + P$. So the two answers in (i) are consistent.
- For $a = \frac{1}{2}$, $h = \frac{2}{5}$, $\mu = \frac{3}{4}$ the four thresholds above were computed exactly, and the comparisons in (ii)–(iv) were confirmed by computer on many random choices of a, h and μ.

Common slips

- In (iii), comparing only slipping with toppling of both cubes, and never showing that the upper cube cannot topple first.
- Taking moments about a point on the ground for the upper cube, which brings in the friction F_1 at the interface; moments about X avoid it.
- Using h instead of $1 + h$ as the height of P above the ground in the whole-system moment equation.

Beyond STEP

Each failure mode here is an inequality on the forces, and equilibrium survives as long as all of them hold at once; the first one to fail is the one with the smallest threshold. This idea of a set of linear constraints with a "binding" one is the heart of the limit analysis used for masonry and stacked structures in engineering, and of linear programming more widely. For a careful treatment of friction, torque and the statics of rigid bodies, see Kleppner and Kolenkow, *An Introduction to Mechanics*.

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